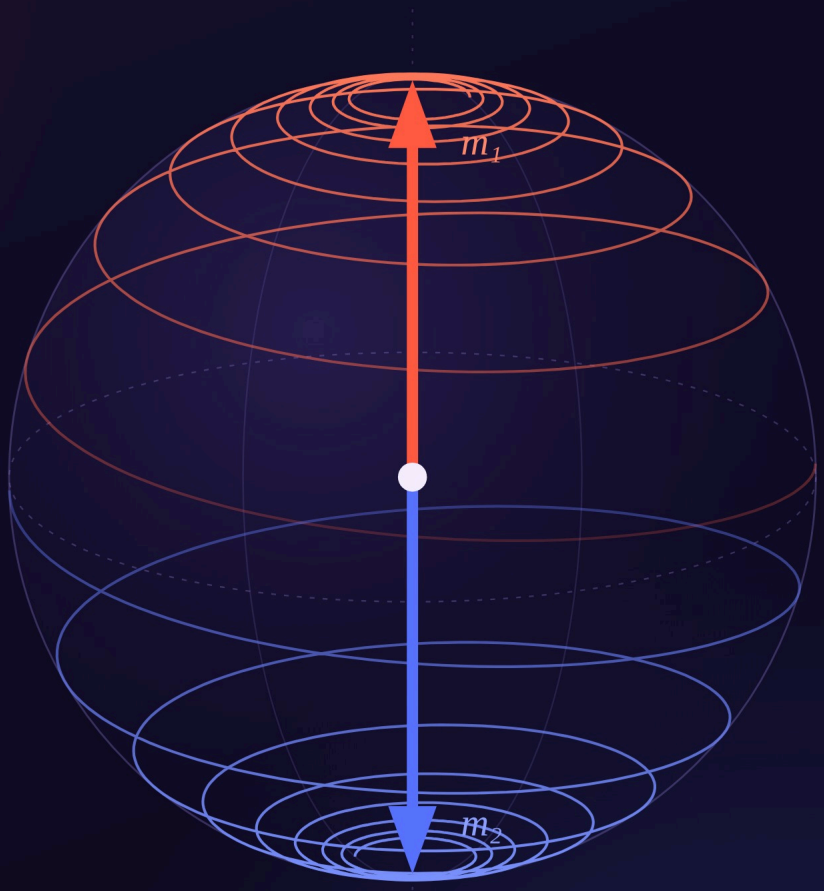
spinmate



USER GUIDE

Spintronics simulation, by conversation.

An AI agent that validates your setup with the solver's own parser, runs two-sublattice macrospin simulations and explains the results from the actual magnetization dynamics, with figures and reports ready to share.

Validated solvers
Macrospin LLG on 9 analytic cases; 2D micromagnetics for skyrmions; mumax3 in

Claude, OpenAI, Gemini
Routed per request by capability and cost

Any agent (MCP)
Web app, Claude Code, Claude Desktop

Runs locally
Docker or Windows app; your keys, your data

Hermes · Dionysus · Zeus · Apollo · Athena

spinmate

User Guide

Version 2.0 · free and open source (MIT) · github.com/Michaellianeris/Spinmate-AI-Agent

An AI agent for spintronics simulation. Describe the setup in plain language or provide a parameter file. SpinMate validates the input with the solver's own parser, runs two-sublattice macrospin simulations, and explains the results from the actual magnetization dynamics, with figures and reports ready to share.

At a glance

- Chat-driven runs, sweeps and threshold searches on the two-sublattice macrospin (LLG) solver
- Answers grounded in computed metrics: switches, settling, frequencies
- **New in 2.0:** an MCP server, so Claude Code, Claude Desktop or any agent uses the same tools
- **New in 2.0:** a physics validation suite against analytic results, SpinMate's own built-in 2D micromagnetic engine (skyrmions, domain walls) and a mumax3 engine plug-in
- Claude, OpenAI and Gemini behind four personas, chosen per request by capability and price
- Figures (IEEE, Nature), reports with LaTeX math, a searchable run catalog
- Six interface languages; runs locally with Docker or a Windows app

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1 Getting started

Docker (Linux, macOS, Windows/WSL)

```
cp .env.example .env      # add ANTHROPIC_API_KEY, OPENAI_API_KEY and/or GEMINI_API_KEY
docker compose up -d      # then open http://localhost:8421
```

Windows desktop app

Run `SpinMate.exe`; it starts the server and opens the browser. Runs are kept in `%LOCALAPPDATA%\SpinMate\data` and the catalog is SQLite.

First start

1. **Athena**, the owl, asks your name and, optionally, your email. The email creates an account and, if a mail server is set up, brings this guide as a welcome email; otherwise she gives you the download link.
2. **API key:** Settings (gear) → paste an Anthropic, OpenAI and/or Gemini key (Gemini keys: aistudio.google.com/api-keys). A pasted key stays in that browser tab; a key in `.env` serves everyone on the server.
3. **Ask** in the message box, or pick a card under *Find a skill* (FM → AFM, FM → FM, Néel and Bloch skyrmions, a hopfion, skyrmion motion, a

synthetic-AFM skyrmion, a skyrmion at 300 K, spin waves, FeRh heating).

> No key? *Run setup* (play icon) runs a preset or your parameter file directly; only the chat needs a model.

2 Your first simulation

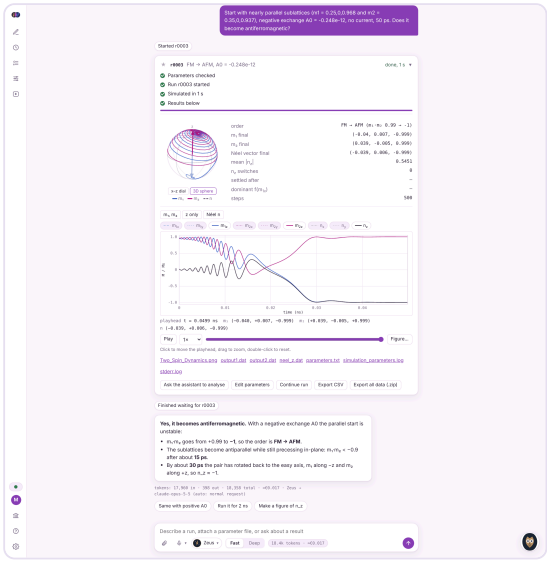
Describe the starting state, exchange, fields, current or temperature, and the duration (SI units, currents in A/cm²).

> *Start with nearly parallel sublattices (m1 = 0.25,0,0.968 and m2 = 0.35,0,0.937), A0 = −0.248e-12, 50 ps. Does it become antiferromagnetic?*

> *Start antiferromagnetic with a positive exchange A0 = 0.248e-12, 120 ps. Does it end ferromagnetic?*

> *Apply a DC spin-orbit current of 100e6 A/cm² for 2 ns to the default AFM state. What happens?*

> *Heat a FeRh layer through its transition: positive A0, 370 K ± 60 K at 0.25 GHz, 2 ns.*



Question, run card and answer with next steps.

The parameters are checked with the solver's own parser before anything runs; a suspicious value (for example `A0 = 0.248` without `e-12`) is pointed out instead of run. Large requests (many runs or long runtimes) wait for your **Run it** button. The answer ends with suggested next steps as buttons.

What happens behind one question

1. The router picks a model for the persona and the request.
2. The model calls `validate_parameters`, then `start_simulation`.
3. The run queue starts the solver in its own process; the card shows live progress.
4. `wait_for_run` returns the summary metrics; the model writes the answer from them.

3 Reading a run card

Field	Meaning
order	Initial and final order from $m_1 \cdot m_2$: AFM below −0.9, FM above +0.9, otherwise canted.
m_1, m_2, Néel final	Final unit vectors and the Néel vector $n = (m_1 - m_2)/2$.
n_z switches	Real reversals of n_z (from below −0.5 to above +0.5 or back) and the time of the first; ringing around zero is not counted.
settled after	When m_1 and m_2 stop changing within 2 %; "–" if they did not settle within the run.

dominant f	Main frequency of m_{1x}, reported only if at least five periods fit in the second half of the run.

The **sphere** shows the trajectories of m_1 (blue) and m_2 (magenta) in 3D or as an x–z dial. The **chart** shows m_1/m_2, z-components only, or the Néel vector: drag to zoom, double-click to reset, click to move the playhead; **Play** animates the run. The card links the raw files and exports CSV or a zip of all data.

4 Fast, Deep and next steps

Fast uses the default parameters, changes only what you name and answers in a few lines. **Deep** works from the full parameter set or your file, states the caveats that apply and gives a quantitative explanation. Both end with up to three suggested follow-ups. Requests estimated above `CONFIRM_MINUTES` or `CONFIRM_RUNS` pause for **Run it**.

5 Models and personas

Under the message box you pick a persona, not a model. Each persona groups models of one level from all three providers.

Persona	Best for	Typical models
Hermes	Single runs, presets, short answers	Claude Haiku · GPT nano / mini · Gemini Flash-Lite
Dionysus (default)	Everyday runs with a clear explanation, sweeps, comparisons	Claude Sonnet · GPT-5.x · Gemini Flash
Zeus	Deep mode, goal searches, careful curve reading, reports	Claude Opus · OpenAI o-series · Gemini Pro
Apollo	New or special models as they appear	Claude Fable / Mythos · new Gemini lines · new models
Athena	The how-to guide: where things are, what buttons do	Selected provider · built-in answers offline

How the model is chosen

On *Auto* each request is scored for every model of the persona whose key is present:

```
score = capability + bonus(strongest of the persona) − λ(persona, request) × log10(price / cheapest) − shortfall
```

Capability comes from the model family and generation; price from list prices; λ is largest for Hermes and smallest for Zeus and shrinks for demanding requests (Deep mode, sweeps, comparisons, reports, long messages). So the strongest model wins unless a cheaper one is clearly enough. There is no fixed provider order; on an API error the next model by score is used. Pro models run only when pinned (Settings → *Model behind each god*). Every answer shows the model, the reason and the cost in €, converted with the daily ECB reference rate.

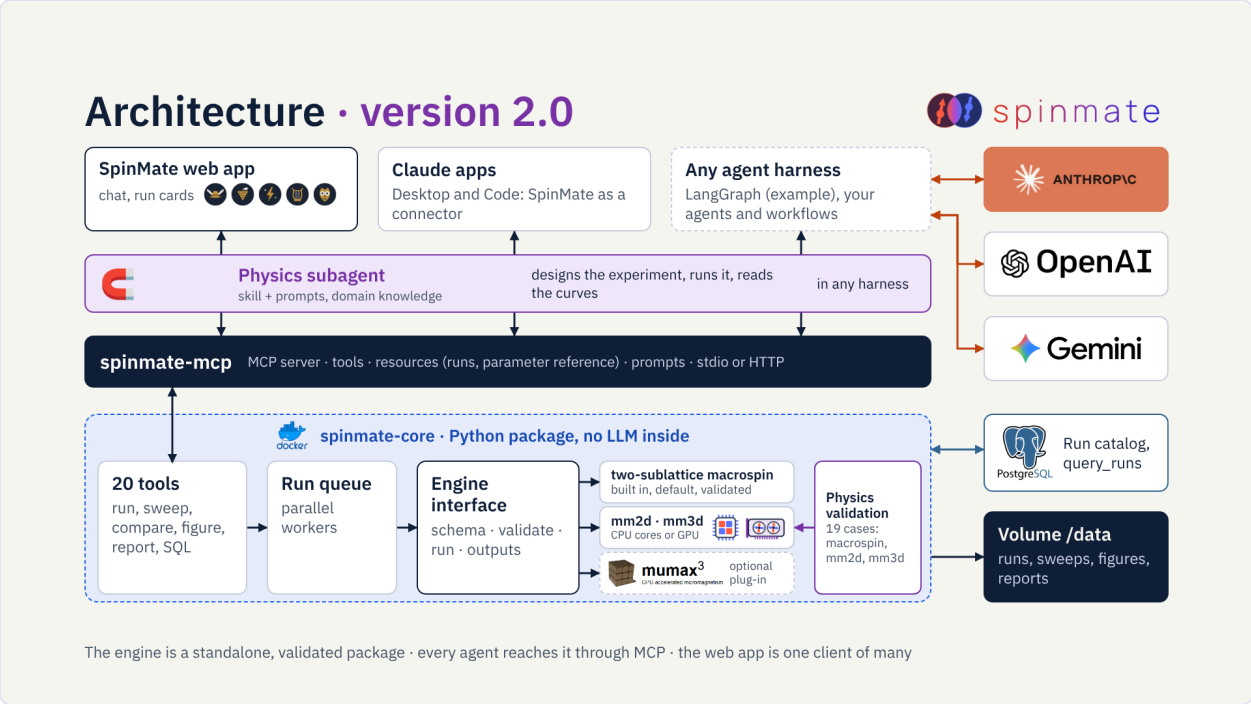
6 The tools

Group	Tools
Inputs	`list_input_files`, `read_input_file`, `save_input_file`, `validate_parameters`, `list_engines`

Runs	`start_simulation`, `wait_for_run`, `get_run`, `list_runs`, `cancel_run`
Sweeps	`start_sweep`, `wait_for_sweep`, `get_sweep`
Continuation	`continue_run`, `combine_runs`
Analysis & output	`compare_runs`, `make_figure`, `make_report`, `query_runs` (SQL on the catalog)

The same 19 tools serve the chat and every MCP client (chapter 8). `query_runs` appears only when the catalog is on. `start_simulation` and `start_sweep` stop large requests: in the chat they wait for **Run it**, over MCP they return `needs_confirmation` and run only when called again with `confirmed=true`.

7 Architecture



Version 2.0: every client reaches the same MCP server, on top of spinmate-core and its engines.

Component	Role
Browser (`web/index.html`)	Chat, run cards, sphere and charts, Runs panel, figure and report cards, voice input, six languages. Conversations are kept in the browser.
FastAPI server (`app/server.py`)	REST endpoints for runs, sweeps, figures, reports, accounts, transcription; streams the agent's events to the page; serves the MCP server at `/mcp`.
Agent loop (`app/agent.py`)	Provider-neutral tool loop for Claude, OpenAI and Gemini, with prompt caching, token budget and cost accounting. It calls the tools through MCP (`app/mcp_client.py`), like any other client.
Model router (`app/router.py`)	Scores the persona's models on capability and price for each request.
MCP server (`spinmate_mcp/`)	Tools, resources (parameters, physics guide, validation, runs, examples) and prompts, over stdio or HTTP. `physics.md` is the physics guide shared by the server, the chat and the Claude Code subagent.
spinmate-core (`spinmate_core/`)	The engine without an LLM: tools, run queue, sweeps and continuation, catalog, figures, reports, validation suite and a command line (`python -m spinmate_core`).

Engines (`spinmate_core/engines/`)	One interface: `parameter_schema`, `validate`, `run_command`, `outputs`. macrospin and mm2d are built in; plug-ins such as mumax3 register under `spinmate.engines`.
Run catalog	Postgres in Docker, SQLite in the desktop app: one row per run, queried with read-only SQL.
Data volume	`/data`: runs, sweeps, figures, reports, validation, accounts, exchange-rate cache.

8 From Claude and other agents (MCP)

SpinMate is also an MCP server (Model Context Protocol), so Claude Code, Claude Desktop or any MCP agent can design, run and read simulations with the same tools, runs and queue as the web app. The plug icon in the left bar shows these commands for your server, with copy buttons.

Claude Code

With the Docker app running:

```
claude mcp add --transport http spinmate
http://localhost:8421/mcp
```

The repository also brings a physics subagent (`.claude/agents/spinmate-physicist.md`) and a skill (`.claude/skills/spinmate-physics`), used automatically when you open Claude Code in the repository.

Without Docker

```
python -m spinmate_mcp          # stdio
python -m spinmate_mcp --http   # port 8422
```

Claude Desktop

Settings → Developer → Edit config (`claude_desktop_config.json`):

```
{"mcpServers": {"spinmate": {
  "command": "docker",
  "args": ["compose", "-f",
"/path/to/Spinmate-AI-Agent/docker-
compose.yml",
           "exec", "-T", "sublattice-agent",
"python", "-m", "spinmate_mcp"]}}}
```

Links in the results (figures, reports) point to `APP_URL`, by default `http://localhost:8421`.

The server offers	Contents
Tools	The 20 tools of chapter 6; `get_run` and `make_figure` return the plot as an image.
Resources	`spinmate://parameters`, `physics/guide`, `physics/scales`, `validation`, `runs`, `runs/{id}`, `runs/{id}/parameters.txt`, `examples/{name}`
Prompts	`design_experiment` (a question → runs → interpretation), `find_threshold` (coarse and fine sweeps), `interpret_run`

> *Use spinmate to find the smallest DC SOT current that makes the Néel vector rotate.*

LangGraph workflows (example)

`examples/langgraph` drives SpinMate from a LangGraph graph through the MCP server: a threshold search (plan → sweep → wait → analyse → narrow, up to 4 rounds) with its state checkpointed in SQLite, so a search over long runs goes on after a restart, and with large sweeps waiting for your yes (`interrupt()`). It runs in its own environment:

```
python -m venv .venv-langgraph && . .venv-langgraph/bin/activate
pip install -r examples/langgraph/requirements.txt
cd examples/langgraph
python threshold.py --parameter A0 --lo 0 --hi 1e-14 --criterion order_final=FM --lines="--t=2e-
9" --thread a0
```

It finds the AFM/FM boundary A0 = 1.735e-15 J/m (theory $|Ku|\ l^2/2$ = 1.7395e-15) in three rounds of six runs. Run the same command again with the same `--thread` to resume an interrupted search.

9 Engines: macrospin, mm2d, mm3d and mumax3

Three engines are built in: macrospin (two sublattices), mm2d (2D micromagnetics: skyrmions, domain walls, spin waves) and mm3d (3D micromagnetics: hopfions, bobbers, torons). mumax3 or any other engine is an optional plug-in, added only if you install it. Every run records the engine that produced it. `list_engines` shows what is installed and the parameters of each; in the chat simply ask for a skyrmion or for mumax3, over MCP pass `engine="mm2d"` or `engine="mumax3"`. **Run setup** (Deep mode) has an **Engine** menu; an engine that cannot run here (for example mumax3 without a GPU) is marked “not installed”.

Engine	What it computes	Notes
macrospin (default)	Two coupled LLG equations for the sublattices m_1 and m_2: exchange A0, anisotropy Ku, Zeeman field, spin-orbit torque, thermal noise, time-dependent drives.	Built in; sweeps, continuation and combination of runs.
mm2d (built in)	SpinMate's own 2D micromagnetics of a ferromagnetic film one cell thick: exchange, perpendicular anisotropy, interfacial or bulk DMI, Zeeman and the demagnetising field, local or long-range by FFT (`demag=full`). Finds skyrmions, bubbles, skyrmionium, walls and maze domains, or runs LLG dynamics.	Part of SpinMate: CPU only, nothing to install; a 128 × 128 film relaxes in seconds.
mm3d (built in)	SpinMate's own 3D micromagnetics of a chiral magnet slab or nanodisk with the top and bottom surfaces pinned along +z: exchange, bulk or interfacial DMI, anisotropy, field. Relaxes hopfions (H = 1, 2), skyrmion tubes, chiral bobbers and torons; gives the Hopf index and finds Bloch points.	CPU only; the default hopfion takes about a minute. No demagnetising field.
mumax3 (optional plug-in)	Finite-difference micromagnetics of one ferromagnet on a grid, on the GPU. Input as SpinMate lines (turned into an `.mx3` script) or a complete `.mx3` script; the card shows ⟨m⟩(t).	Needs an NVIDIA GPU and the mumax3 binary in `MUMAX3_BIN` ; without it validation warns. Single runs only.

```
--Nx=128
--Ny=128
--Msat=800e3
--Aex=13e-12
--alpha=0.02
--B_ext=0,0,0.1
--t=2e-9
```

Cross-check with mumax3. Ask *"cross-check this skyrmion with mumax3"* (tool `compare_with_mumax3`): the same mm2d input runs on both engines from the very same starting state, written as an OVF file that mumax3 loads. Grid, cells, material, field, DMI and mask (`ImageShape`) carry over; mm2d's local thin-film demag becomes no demag with $K_{u1} = K_u - \mu_0 M_s^2/2$ in mumax3 (the same energy, exactly), and `demag=full` keeps mumax3's demag. When both are done, a card shows the two final textures, the angle between them and a table of checks with tolerances: Q within 0.1, radius 5 %, energy 2 %, rms angle 5°, and for dynamics ⟨m⟩(t) within

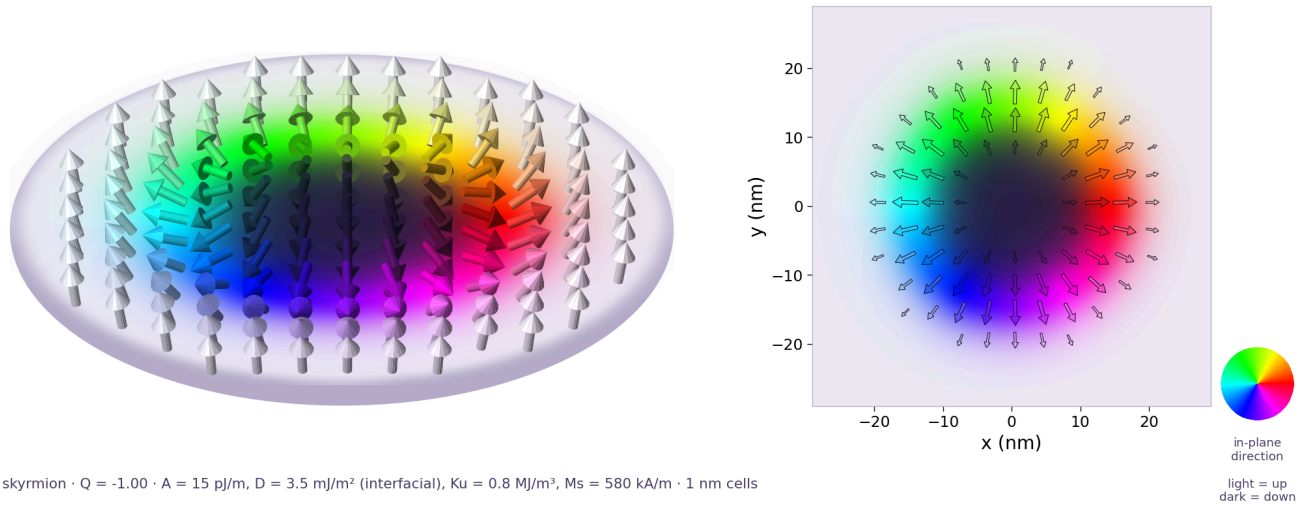
0.02. Not for currents, temperature, antennas or two-sublattice films. The chain is tested with a stand-in for mumax3; it needs a GPU with mumax3 for real.

> mumax3 is free software under the GNU GPL v3; SpinMate starts it as a separate program and stays MIT. If you publish results obtained with it, cite Vansteenkiste et al., *The design and verification of MuMax3*, AIP Advances 4, 107133 (2014). Outside Docker: `pip install -e plugins/spinmate-mumax3`.

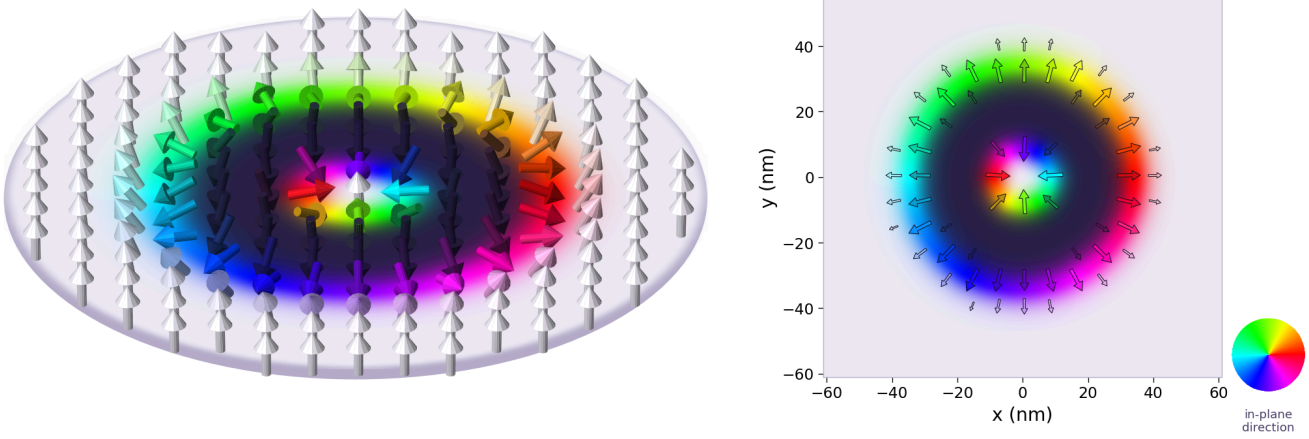
2D micromagnetics: skyrmions, their motion and spin waves with mm2d

Choose **mm2d** in the Engine menu of Run setup (Deep mode), or ask in the chat, for example “relax a Néel skyrmion with D = 3.5 mJ/m²” or “send 80 GHz spin waves from a stripe antenna”. `mode=relax` lowers the energy from a seed (`texture=skyrmion`, `skyrmionium`, `domain_wall`, `uniform`, `random`) to the nearest stable state; `mode=dynamics` relaxes first and then solves LLG in time, with an optional field pulse `B_pulse` for `t_pulse`, and reports the main frequency.

```
--mode=relax
--texture=skyrmion
--Aex=15e-12
--Ku=0.8e6
--Msat=580e3
--D=3.5e-3
--Nx=128
--Ny=128
```



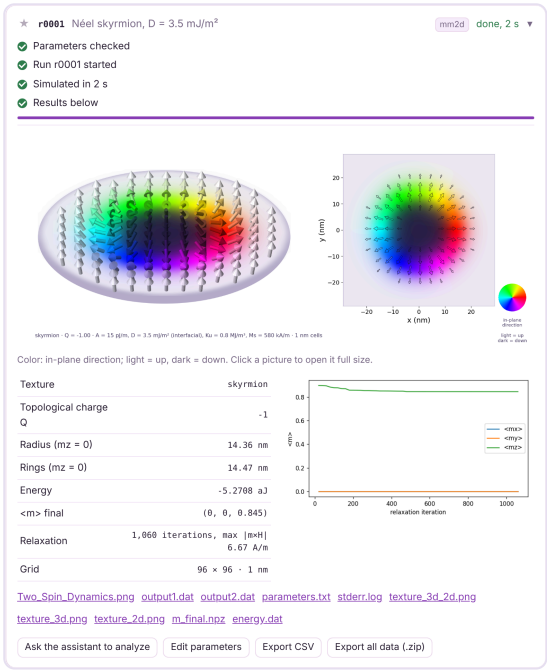
A relaxed Néel skyrmion: tilted view with 3D arrows (left) and top view with nm axes (right). Colour: in-plane direction; light = up, dark = down. Q = −1, radius 14.4 nm.



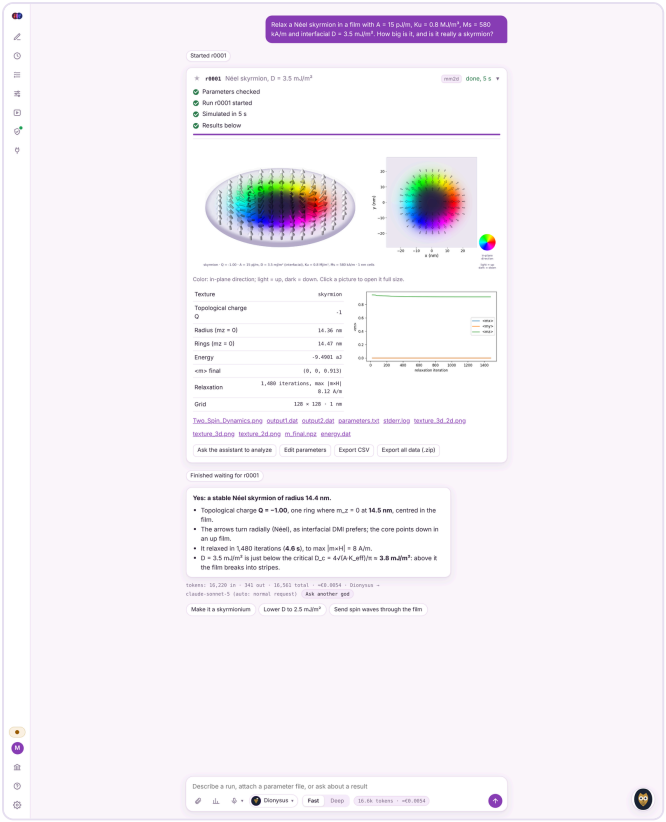
Skyrmionium, a skyrmion inside a skyrmion: Q = 0, rings at 8.5 nm and 34.4 nm.

The run card of an mm2d run shows the final texture in both views, then a table with the texture it found, the topological charge Q, the radius or the rings where $m_z = 0$, the energy and how far the relaxation went, and ⟨m⟩ during the relaxation or in time. `m_final.npz` keeps the full state; click a picture to open it full size.

mm2d check	Expected	Measured
Wall energy, D = 0 / 2 mJ/m²	$4\sqrt{(A \cdot K_{eff})} - \pi\lvert D\rvert$	within 0.3 %
Wall width	$\sqrt{(A/K_{eff})}$	within 1 %
Wall type	Néel (interfacial), Bloch (bulk)	as expected
Kittel frequency	$\gamma_o/2\pi \cdot (H + 2K_{eff}/\mu_o M_s)$	within 0.2 %
Energy, α = 0	conserved	drift < 1e-4
Skyrmion / skyrmionium	Q = −1 / 0	−1.00 / 0.00
Spin wave, 80 GHz	λ = 49.6 nm	49.3 nm
Dispersion f(k)	analytic curve	within 1 %



The run card of an mm2d skyrmion.



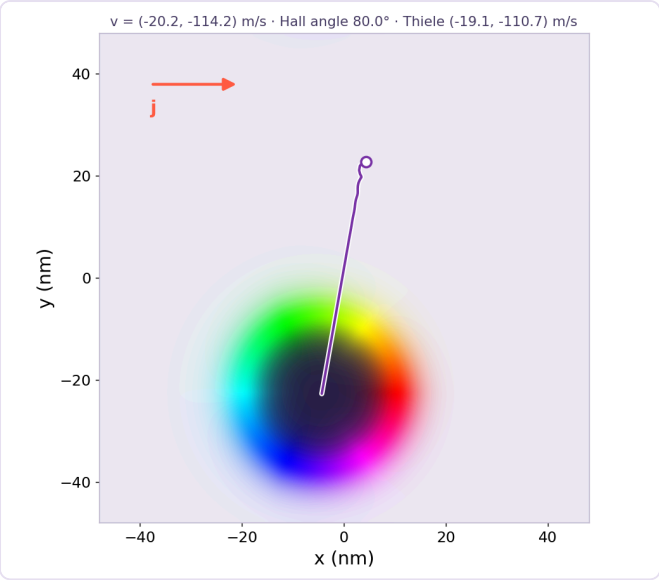
In the chat: a skyrmion from a question.

In the chat: spin waves from a question.

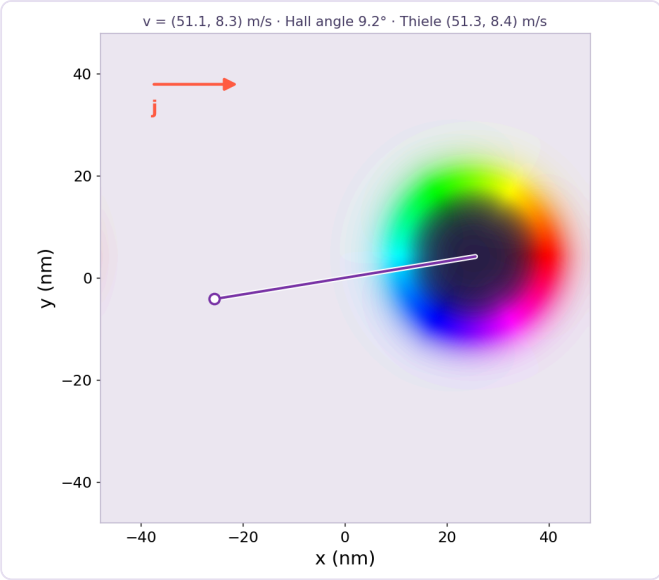
Skyrmion motion with a current

In `mode=dynamics` a current density `j` (A/m², in the plane) moves the texture. `stt_P` turns on spin-transfer torque in the film (Zhang-Li, drift velocity $u = P\,\mu_B\,j/(e\,M_s)$, non-adiabaticity `stt_beta`); `theta_SH` turns on the damping-like spin-orbit torque from a heavy metal under the film ($H_{DL} = \hbar\,\theta_{SH}\,|j|/(2e\,\mu_0\,M_s\,d)$, polarisation $\sigma = z \times j$). The card shows the path of the centre, the velocity, the skyrmion Hall angle and the Thiele prediction for the relaxed texture.

```
--mode=dynamics
--texture=skyrmion
--Nx=96
--Ny=96
--alpha=0.1
--j=1e12,0,0
--theta_SH=0.1
--t=400e-12
```

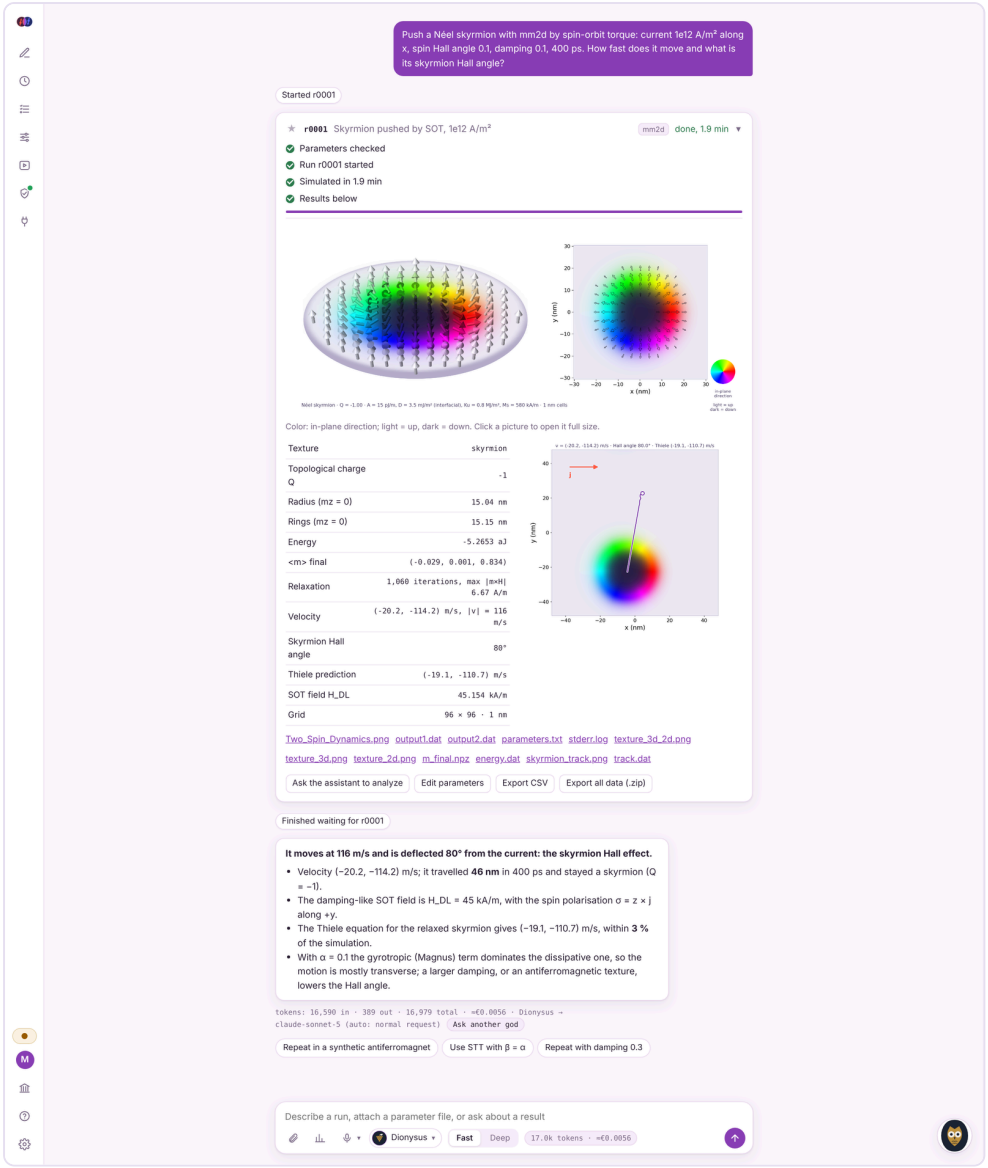


SOT, 1e12 A/m²: 116 m/s, deflected 80° from the current (skyrmion Hall angle); Thiele within 3 %.



STT, u = 50 m/s, β = 0.2, α = 0.1: Hall angle 9.2°; Thiele (51.3, 8.4) m/s.

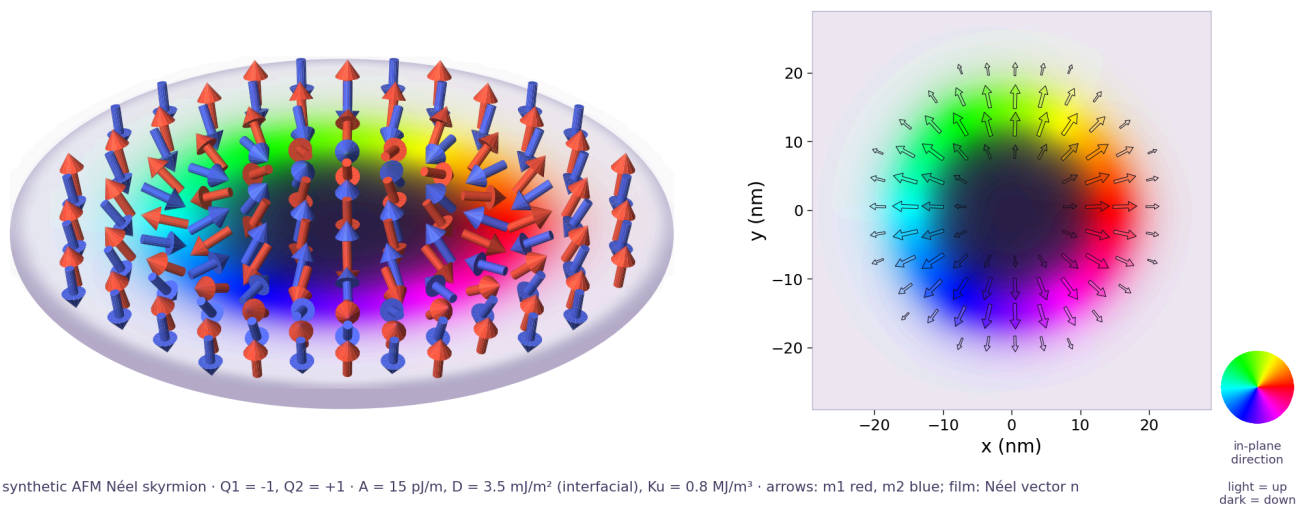
Motion check	Expected (Thiele)	Measured
STT, β = α	v = u = 50 m/s	49.8 m/s, no Hall angle
STT, β = 2α	(51.4, 8.2) m/s	(51.5, 8.1) m/s
SOT, 2e11 A/m²	(−3.8, −22.1) m/s	(−3.8, −22.0) m/s



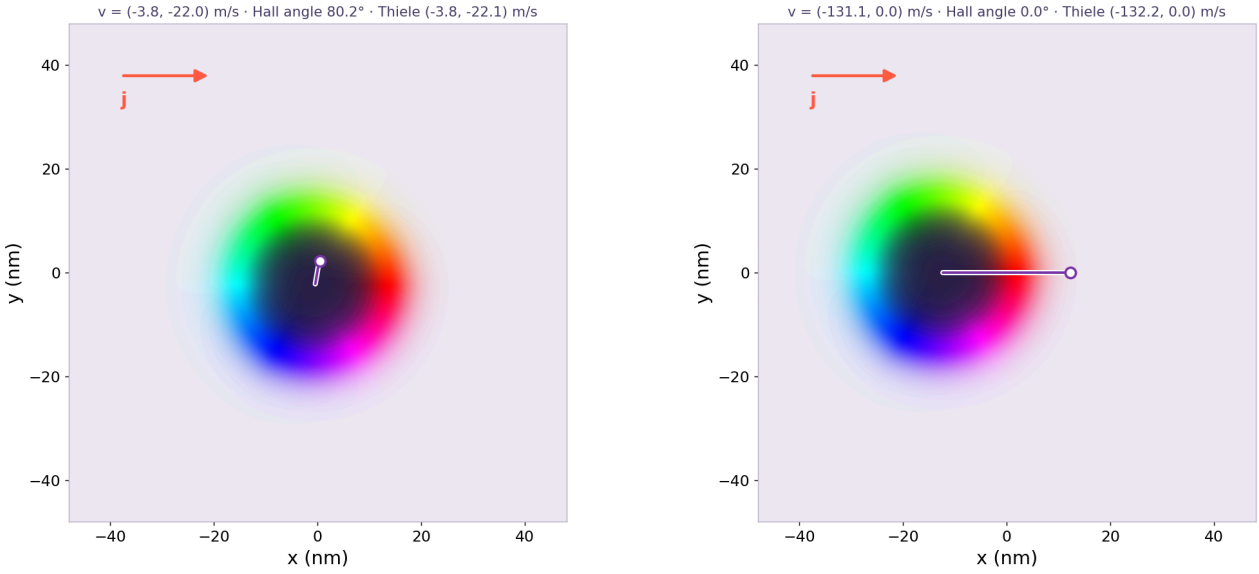
In the chat: how fast a skyrmion moves under SOT.

Antiferromagnetic skyrmions

`lattice=saf` is a synthetic antiferromagnet: two ferromagnetic layers coupled antiparallel through a spacer (RKKY, `J_rkky` < 0, energy $-J_{rkky}\, m_1 \cdot m_2$ per area), each with its own thin-film field, as in Pt/Co/Ru multilayers where such skyrmions are observed at room temperature. `lattice=afm` is an intrinsic, compensated antiferromagnet with two sublattices coupled by `J_afm` and no demagnetising field. The two skyrmions have opposite charges ($Q_1 = -1$, $Q_2 = +1$): their Magnus forces cancel, so under a current they move along the current axis with no skyrmion Hall effect. The pictures show the Néel vector with the arrows of both layers (m_1 red, m_2 blue).

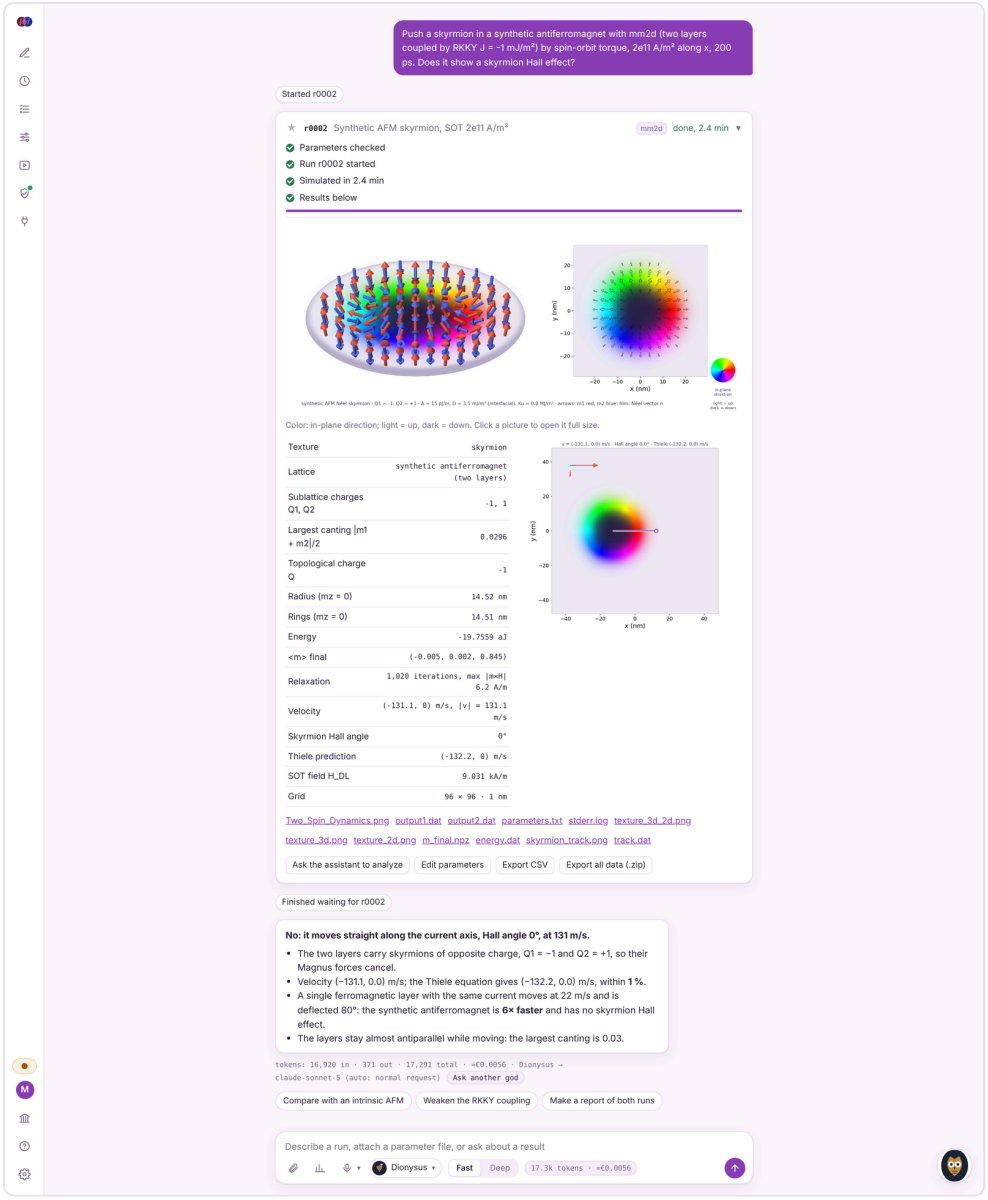


A skyrmion in a synthetic antiferromagnet, J_{rkky} = −1 mJ/m²: R = 14.5 nm, Q_1 = −1, Q_2 = +1.



The same SOT current, 2e11 A/m²: one ferromagnetic layer (left), 22 m/s deflected 80°; the synthetic antiferromagnet (right), 131 m/s along the current.

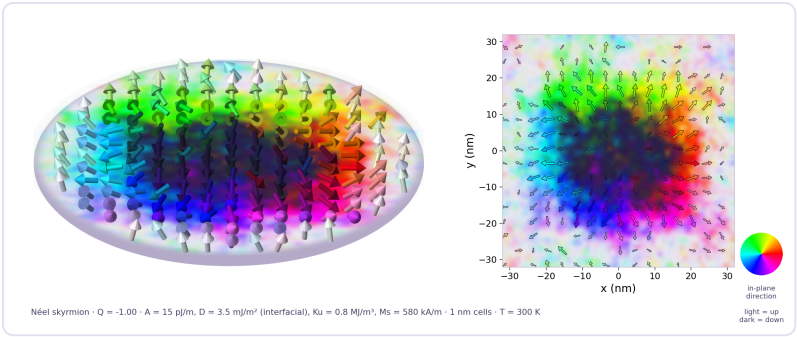
Antiferromagnet check	Expected	Measured
AFM skyrmion profile	ferromagnet without demag	same radius
Skyrmion Hall angle (AFM, SAF)	0	0.0°
SAF velocity, 2e11 A/m²	Thiele 132.2 m/s	131.1 m/s



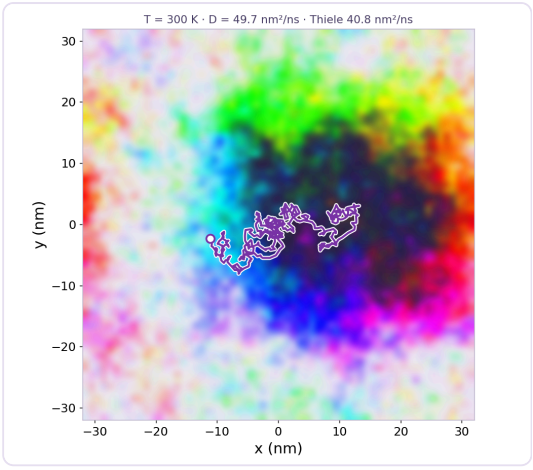
In the chat: does a synthetic-AFM skyrmion show a skyrmion Hall effect?

Temperature

`Temp` (K) in `mode=dynamics` adds the stochastic thermal field of Brown, $\langle H_i(t)H_j(t')\rangle = 2\alpha k_B T/(\mu_o M_s \gamma_o V)\,\delta_{ij}\delta(t - t')$, and integrates with fixed Heun steps (`thermal_dt` , by default 0.15 of the fastest precession period). That is slower than at 0 K, so keep thermal runs to about 64 × 64 cells and 1 ns. Relaxation (`mode=relax`) is an energy minimisation and stays at 0 K, so `Temp` has no effect there and Check says so; a thermal run relaxes at 0 K first and then evolves at T. `start_from=r0003` starts a run from the final texture of an earlier run (same grid and lattice), for example to heat or push a skyrmion that is already relaxed; the card links back to that run. For a skyrmion without current the card gives the diffusion constant from the mean squared displacement, $\langle r^2\rangle = 4D\tau$, next to the stochastic Thiele value; one 1 ns trajectory has a statistical error of about 25 %.



A Néel skyrmion at 300 K.



Its Brownian path over 1 ns: D = 49.7 nm²/ns, Thiele 40.8 nm²/ns.

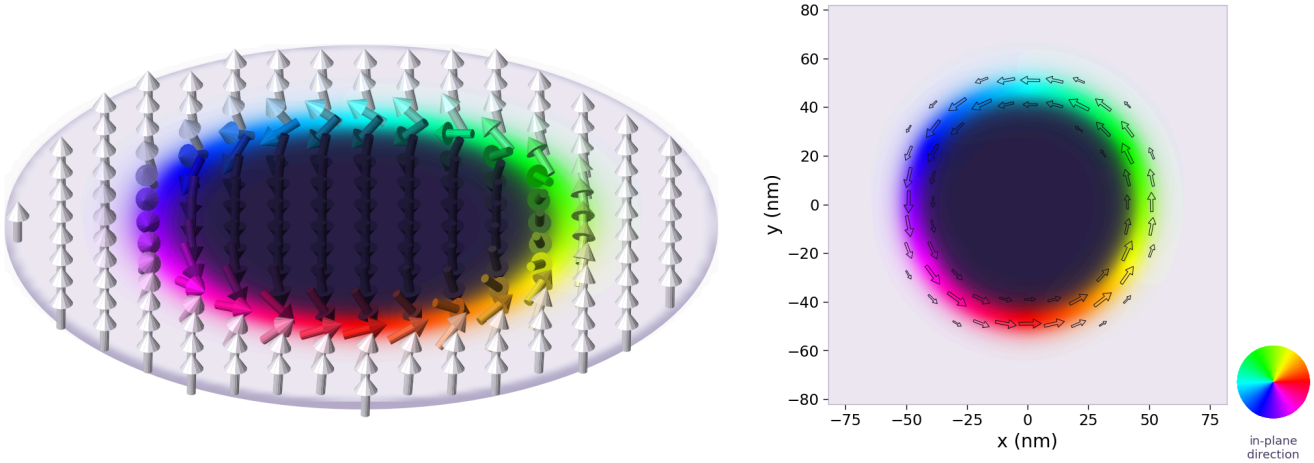
Thermal check	Expected	Measured
Independent spins, 300 K, α = 0.1 and 0.5	Boltzmann $\langle m_z \rangle$ = 0.058	0.058 ± 0.002
Skyrmion diffusion, 300 K, α = 0.3	Thiele 40.8 nm²/ns	49.7 nm²/ns (one trajectory, ±25 %)

In the chat: how far does a skyrmion wander at 300 K?

Long-range demag: bubbles and domains

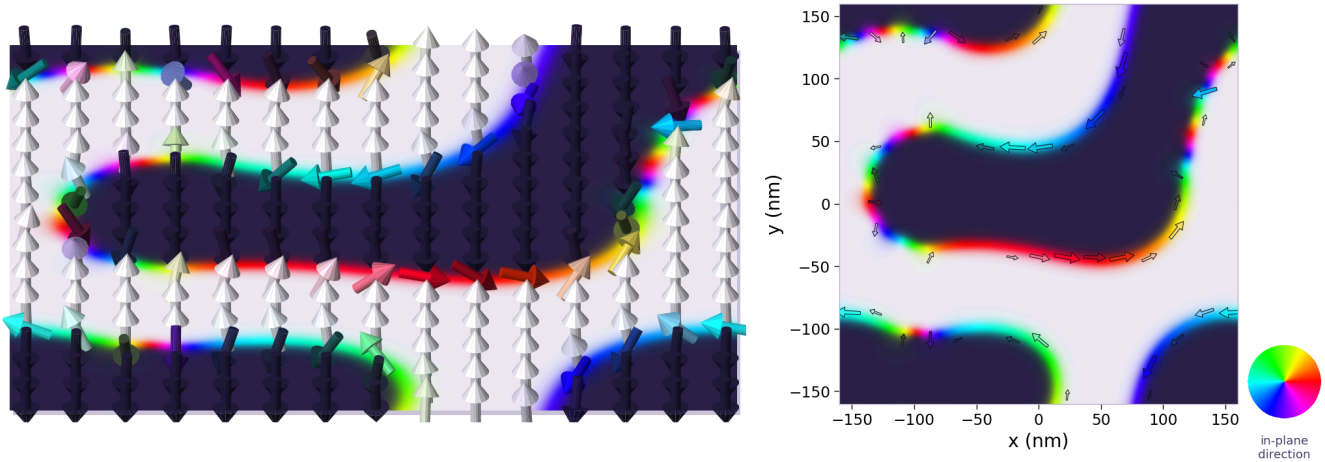
By default the demagnetising field is local, $-M_s m_z$: exact for a uniform film and good for films a few nm thick. `demag=full` computes the stray field of the whole film by FFT with the Newell tensor of a cell `thickness` thick (zero-padded for free edges; for a periodic film with images and the exact infinite-film term). It gives quantitative sizes in thicker films, dipolar bubbles without DMI and stripe or maze domains from `texture=random`, which the local field cannot produce; it costs about 2× (periodic) to 4× (free edges) per step. Check warns when a film is thicker than twice the exchange length and still uses the local field.

```
--texture=skyrmion
--D=0
--demag=full
--thickness=10e-9
--dx=2e-9
--Msat=1e6
--Ku=0.8e6
--B_ext=0,0,0.05
--R0=50e-9
```



bubble · Q = -1.00 · A = 15 pJ/m, no DMI, Ku = 0.8 MJ/m³, Ms = 1000 kA/m · 2 nm cells · 10 nm thick, full demag (FFT)

A dipolar bubble in a 10 nm film, no DMI, 50 mT: R = 47 nm; without DMI the wall turns Bloch.



domains · Q = -3.00 · A = 15 pJ/m, no DMI, Ku = 0.8 MJ/m³, Ms = 1000 kA/m · 2 nm cells · 20 nm thick, full demag (FFT)

Maze domains in a 20 nm film from a random start (`texture=random`, `demag=full`).

Demag check	Expected	Measured

Platelet, uniform M	Aharoni prism factor	within 10^{-6}
Néel − Bloch wall, 0.5 nm film	$\mu_0 M_s^2 d \cdot \ln 2/\pi$ = 0.0466 mJ/m²	0.0464 mJ/m²
Skyrmion radius, 1 nm / 2 nm film	larger than with local demag (14.5 nm)	17.2 nm / 24 nm
Bubble, 10 nm film, 0 / 20 / 50 mT	shrinks with field	93 / 79 / 47 nm

Materials and export

`--material=FeGe` (or `Pt/Co`, `Co/Pd`, `MnSi`, `YIG`) sets M_s, A, K_u, D and the kind of DMI and suggests the cell, grid, field and seed radius that give a skyrmion; in mm3d a hopfion disk. Your own lines win. So *"a skyrmion in FeGe"* runs without numbers: a Bloch skyrmion of R = 37 nm at 0.37 T. MnSi's A and D follow from its 18 nm period and 0.6 T critical field.

Material	M_s (kA/m)	A (pJ/m)	K_u (MJ/m³)	D (mJ/m²)	Source
`Pt/Co`	580	15	0.8	3 interfacial	Sampaio et al., Nature Nanotechnology 8, 839 (2013)
`Co/Pd`	880	10	0.727	2 interfacial	Pollard et al., Nature Communications 8, 14761 (2017)
`FeGe`	384	8.78	0	1.58 bulk	Beg et al., Scientific Reports 5, 17137 (2015)
`MnSi`	163	0.4	0	0.28 bulk	Mühlbauer et al., Science 323, 915 (2009) (period, phase diagram); values derived here
`YIG`	140	3.7	0	0	Klingler et al., J. Phys. D 48, 015001 (2015) (A = 3.7 pJ/m)

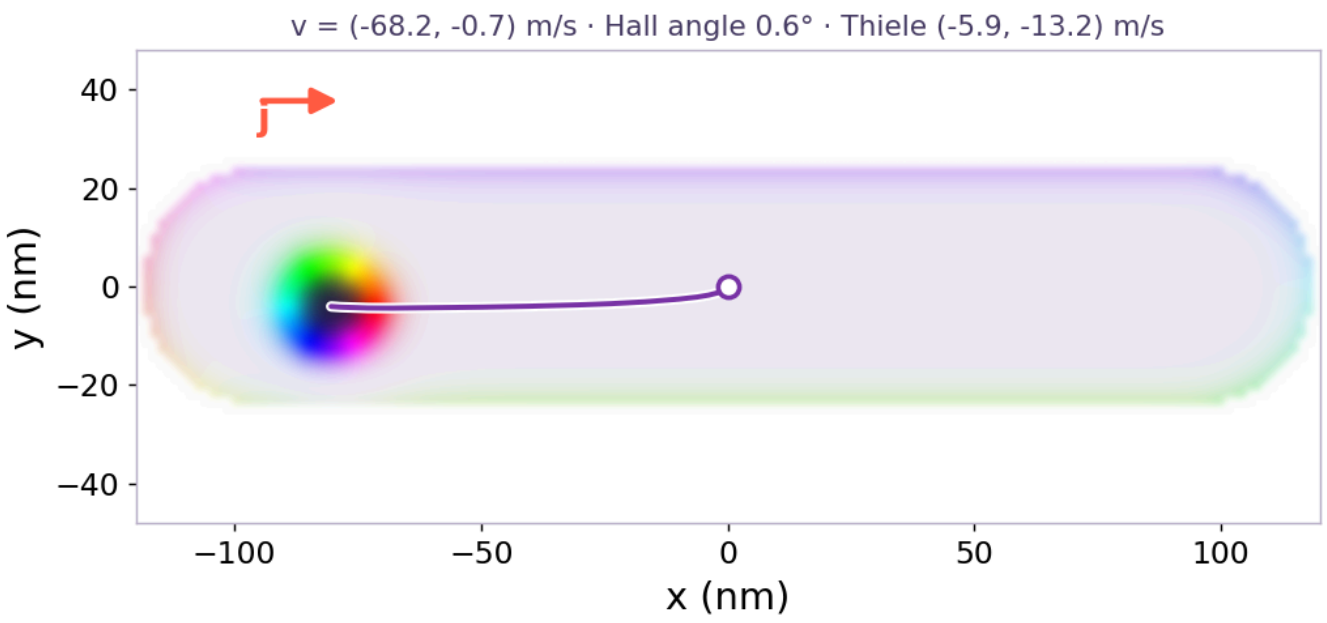
Every mm2d and mm3d run writes `m_final.ovf` (OVF 2.0, binary 4: mumax3 `m.LoadFile`, OOMMF, Ubermag) and `m_final.vtk` (ParaView, VisIt); the run card links them. Empty cells of a mask are zero vectors.

Shapes: disks, strips, racetracks and your own PNG

`mask=disk`, `strip` or `racetrack` cuts the film to that shape. For any other shape draw it dark on a white (or transparent) background, save it as PNG and drop it on the chat with the paperclip: the message says *Attached geometry mask*, and the agent runs with `mask=<name>.png`. The picture is stretched to $N_x \times N_y$ cells, so keep the same aspect ratio (Check warns otherwise). Use `boundary=free`. Ready-made PNGs to edit are in `examples/masks`.

The outline is a free edge with the DMI boundary condition, $2A\,\partial m/\partial n = -D\,(m_z n - (n\cdot m)\,\hat{z})$ for interfacial DMI ($2A\,\partial m/\partial n = D\,n \times m$ for bulk DMI): the magnetisation tilts at the edge, $\sin\theta = D\delta/2A$, and the tilted edge repels skyrmions. The same condition now holds at the free edges of the grid.

```
--mode=dynamics
--texture=skyrmion
--mask=racetrack
--boundary=free
--dx=1.5e-9
--Nx=160
--Ny=64
--alpha=0.1
--j=4e11,0,0
--theta_SH=0.1
--t=1e-9
```

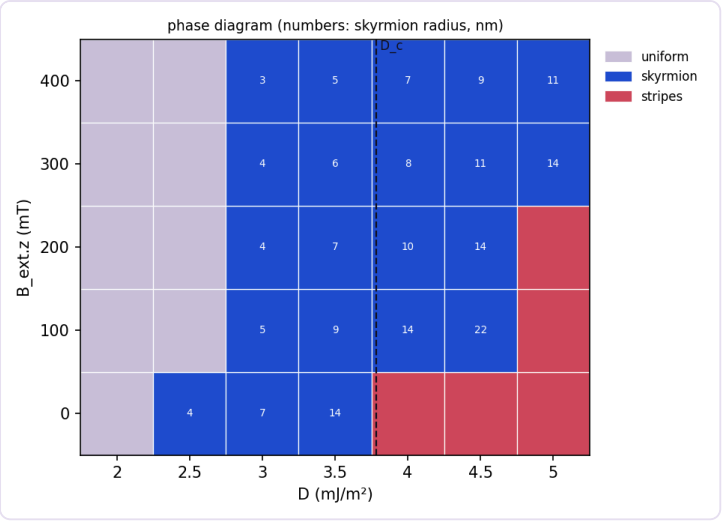


A skyrmion pushed by SOT, 1.5e11 A/m², in a 48 nm racetrack: the tilted edges push back against the skyrmion Hall effect, and the skyrmion runs at 68 m/s along the track, against 14.5 m/s (Thiele) in an open film.

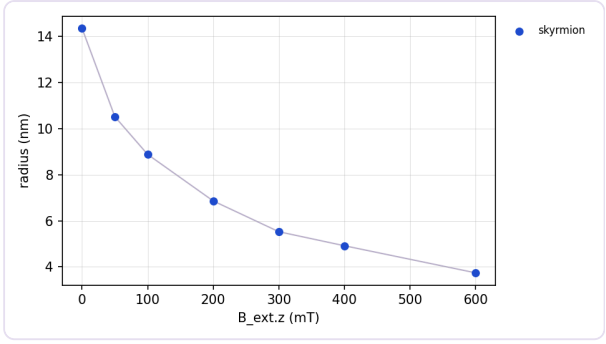
Edge check	Expected	Measured
Tilt at a free edge, interfacial and bulk DMI	sin θ = Dδ/2A = 0.589	0.5887 (within 1 %)
Kind of tilt	Néel (interfacial), Bloch (bulk)	as expected
PNG → mask	same as the built-in shape	identical, top row = +y

Sweeps and phase diagrams

The sweep machinery of the macrospin runs mm2d and mm3d too. `start_sweep` with `engine=mm2d` (or the chat: *"sweep B from 0 to 0.6 T for a skyrmion"*) runs one relaxation per value of any number (`D`, `Ku`, `Msat`, `thickness`, …) or of one component of a vector (`B_ext.z`). The sweep card plots the radius against the parameter, each point coloured by its phase, with a table of the runs. With `parameter2` and `values2` it is a phase diagram, one run per pair: uniform, skyrmion, several skyrmions, stripes, skyrmionium (mm3d: hopfion, tube, bobber, toron). For uniform / skyrmion / stripes start from `texture=skyrmion` with `noise=0.05`, which breaks the symmetry of the seed so an unstable skyrmion can strip out, in a film of at least 128 × 128 cells.



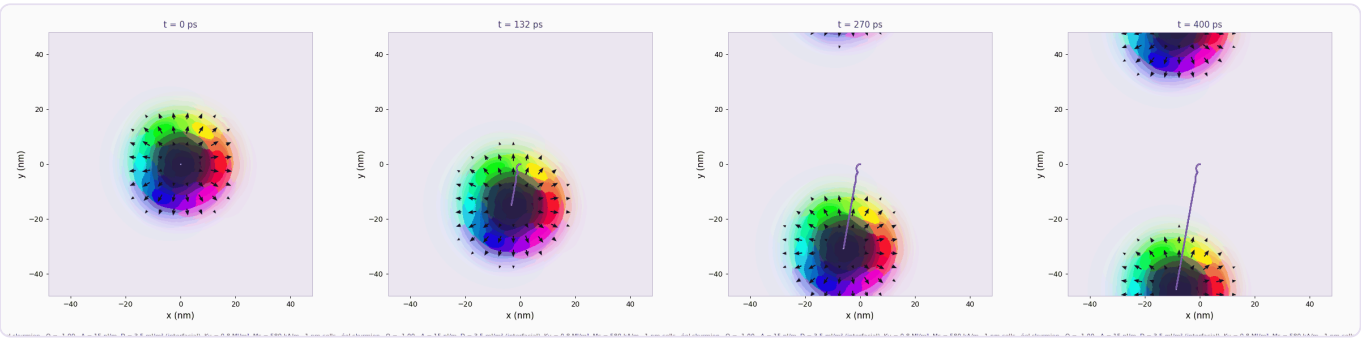
Phase diagram, D × B_z, 35 runs of 128 × 128 cells; the numbers are the skyrmion radius in nm. At B = 0 the stripes start at the dashed line, $D_c = 4\sqrt{(A \cdot K_{eff})}/\pi = 3.78$ mJ/m².



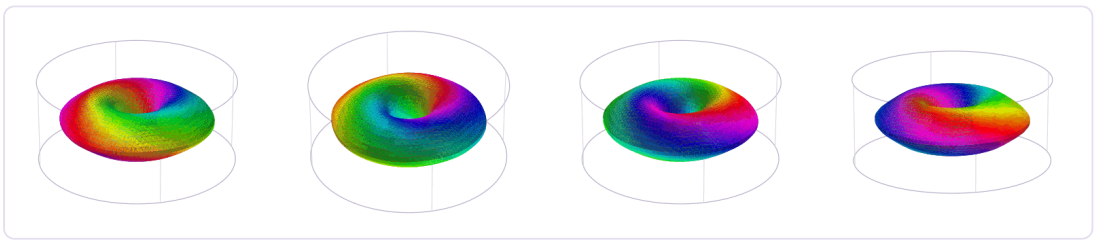
Radius against B_z: 14.4 nm at 0 T, 3.7 nm at 0.6 T.

Animations

A dynamics run with spin waves (`excite=stripe` or `dot`) or with a moving texture (a current, or `Temp`) also writes `animation.gif` , and `animation.mp4` when ffmpeg is installed (it is in the Docker image). Every mm3d run writes `rotate.gif` , the 3D view turning once around z. They play on the run card, with GIF and MP4 downloads; `animation=0` turns them off. Drawing takes about half a minute.



Four frames of the animation of a skyrmion under SOT, 1e12 A/m²: the trail is the path of its centre.

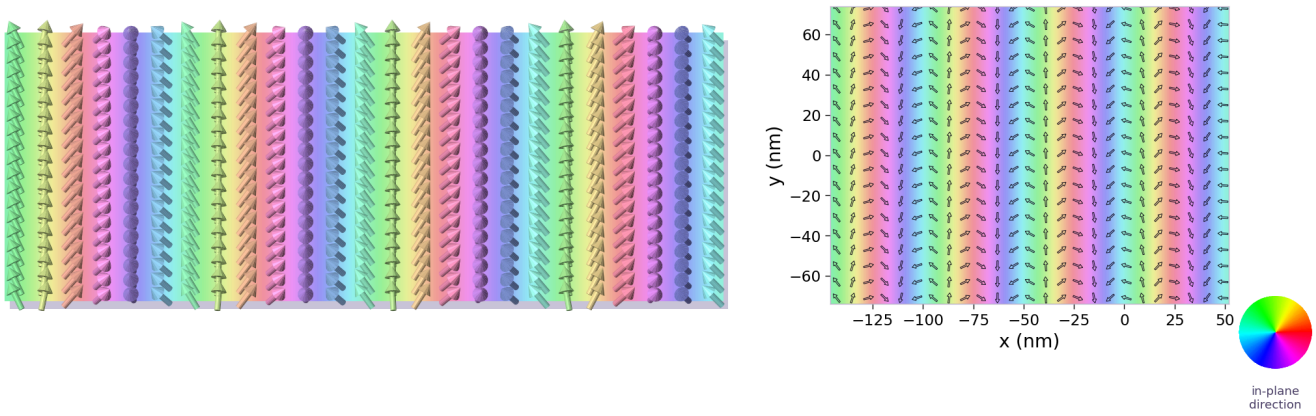


Four frames of rotate.gif of the hopfion.

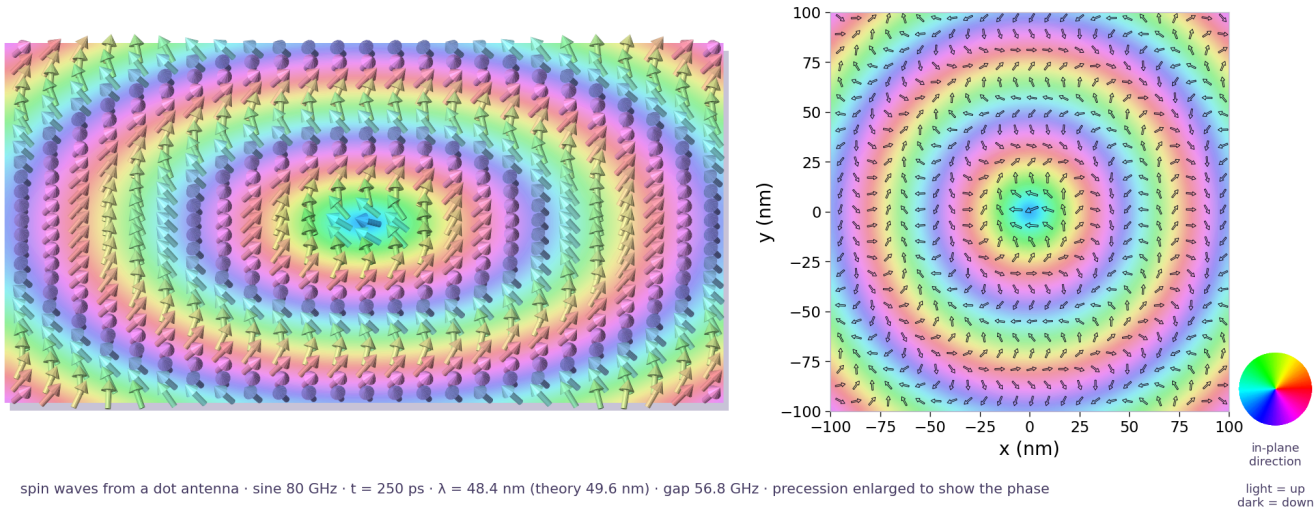
Spin waves

With `excite=stripe` or `excite=dot` the field pulse acts on an antenna instead of the whole film. `pulse=sine` at `f_rf` sends a monochromatic wave: the card gives its wavelength next to the theory. `pulse=sinc` up to `f_rf` excites every wave below that frequency at once and draws the dispersion f(k) with the analytic curve $f = \gamma_0/2\pi \cdot (H + 2K_{eff}/\mu_0 M_s + 2A\,k^2/\mu_0 M_s)$. Set `absorb` to about two wavelengths so the waves do not come back from the edges. Below the gap (here 56.8 GHz) nothing propagates.

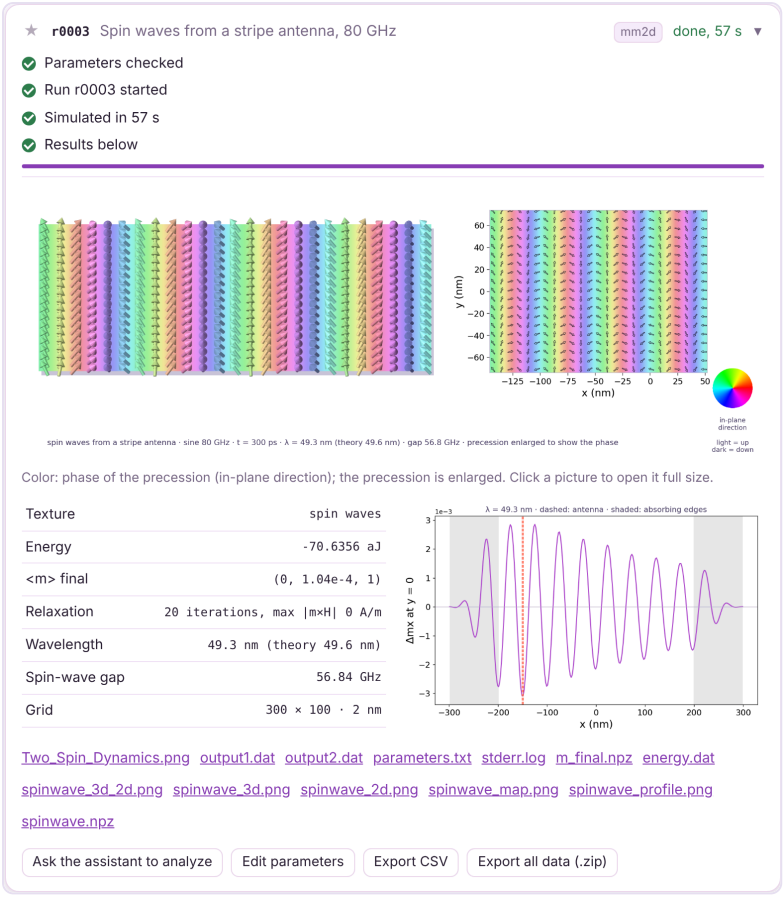
```
--mode=dynamics
--texture=uniform
--Nx=300
--Ny=100
--dx=2e-9
--excite=stripe
--excite_x=-150e-9
--pulse=sine
--f_rf=80e9
--B_pulse=0.02,0,0
--t=300e-12
--t_pulse=300e-12
--table_dt=2e-12
--absorb=100e-9
```



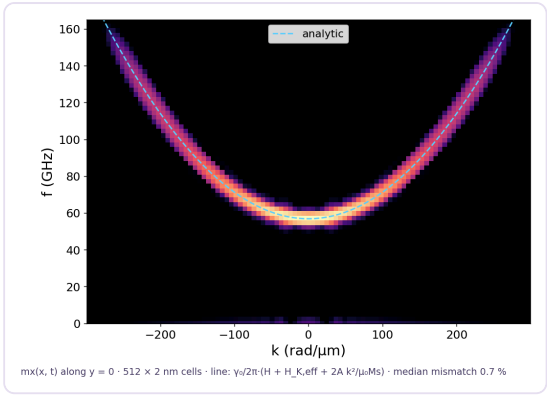
Spin waves at 80 GHz from a stripe antenna, in the same views as a texture. Colour: phase of the precession; the precession is enlarged so it can be seen. λ = 49.3 nm (theory 49.6 nm).



Circular spin waves from a dot antenna.



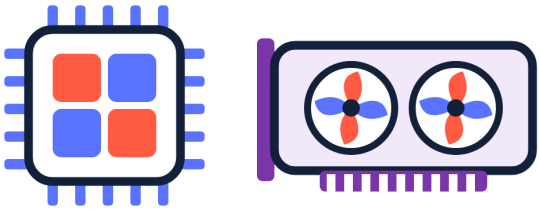
The run card of a spin-wave run: wavelength, gap and the profile along y = 0.



Dispersion from a sinc pulse up to 150 GHz; the dashed line is the analytic f(k).

Limits: the film is one cell through the thickness, so m does not vary along z (mm3d does that). The default demagnetising field is local ($-M_s \cdot m_z$); `demag=full` adds the long-range stray field.

Speed: CPU cores or GPU, and the time step



mm2d and mm3d run on all the cores of your CPU or on an NVIDIA GPU. The physics of a cell (exchange, anisotropy, DMI with its edge condition, demag, field, spin torques, the LLG equation) is written once and compiled for both, so the results are the same: the CPU code reproduces the reference NumPy fields to 10^{-12}.

`backend`	Runs on	When to use it
`auto` (default)	the GPU if there is one, else all CPU cores	always, unless you want to force one
`gpu`	an NVIDIA GPU (CUDA 12), CuPy + numba-cuda	large films, long runs; falls back to the CPU with a note if no GPU is visible
`cpu`	all CPU cores (numba)	any computer; about 8× the NumPy code on 2 cores, more with more cores
`numpy`	one core, the reference code	to compare, or where numba is missing

Where did my run go? The engine menu of Run setup says what `auto` means on this computer (*backend auto = GPU (...)* or *CPU, numba on N cores*). The run card shows *Ran on* and the number of time steps with their mean length; when `gpu` was asked for but not available, Check warns and the card says why. `demag=full` runs on the CPU for now (its FFT is not on the GPU yet).

Using the GPU. You need an NVIDIA GPU and a current NVIDIA driver on Windows; in WSL `nvidia-smi` must list the card, and Docker Desktop must use the WSL 2 backend. Then start SpinMate with the GPU compose file:

```
nvidia-smi
docker compose -f docker-compose.yml -f docker-compose.gpu.yml up -d --build
```

This builds the image with CuPy and numba-cuda (`requirements-gpu.txt`) and gives the container the GPU. If `nvidia-smi` fails there is no usable GPU: start as usual (`docker compose up -d --build`) and the runs use all CPU cores. Outside Docker: `pip install -r requirements-gpu.txt` . The first CPU run after an install compiles the kernels (about 30 s); the Docker image does that at build time.

The time step. Dynamics use the adaptive Dormand–Prince 5(4) method with the error per step `rk_tol` . In practice the exchange of the smallest cells sets the step: the fastest precession on the grid, $\omega_{max} \propto A/(M_s\, dx^2)$, limits it to about $1.5/\omega_{max}$. Temperature uses fixed Heun steps of $0.15/\omega_{max}$. Check gives the number of steps and the time before the run starts, and for runs over half an hour suggests what to change.

Cell size	Time step	Steps per ns
0.5 nm	9 fs	106,000
1 nm	35 fs	28,000
2 nm	123 fs	8,100
4 nm	370 fs	2,700

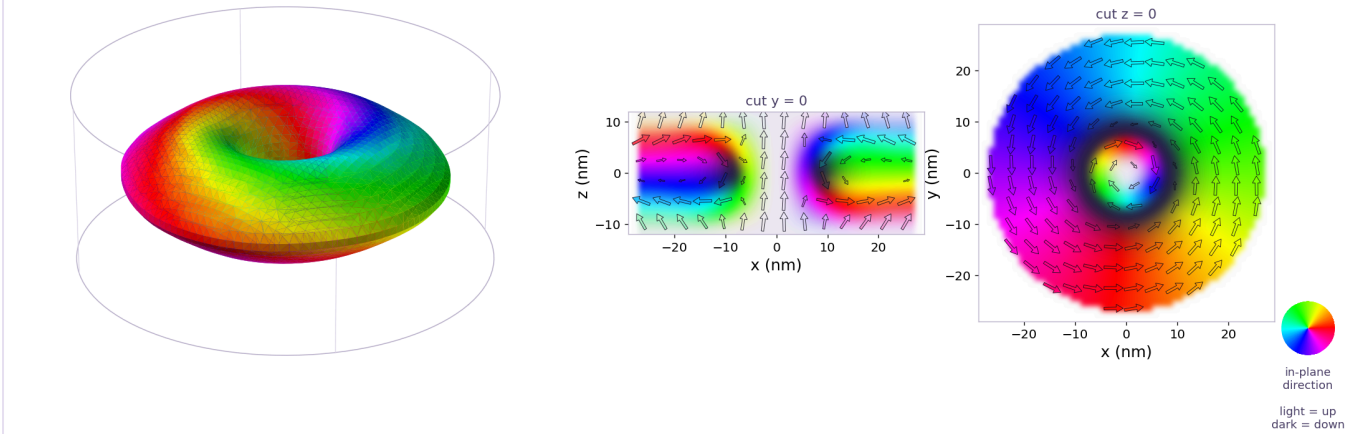
For a Pt/Co-like film (A = 15 pJ/m, M_s = 580 kA/m). On a 2-core machine a 128 × 128 film with 1 nm cells takes 3 ms per step on the CPU backend (25 ms with NumPy), about 1.4 min per ns.

> Long runs (10–100 ns): use cells as large as the textures allow (the wall width $\sqrt{(A/K_{eff})}$ should span 2–3 cells: 2 nm cells give four times fewer cells and four times longer steps), set `table_dt` to 10 ps or more and `animation=0` , and prefer the GPU.

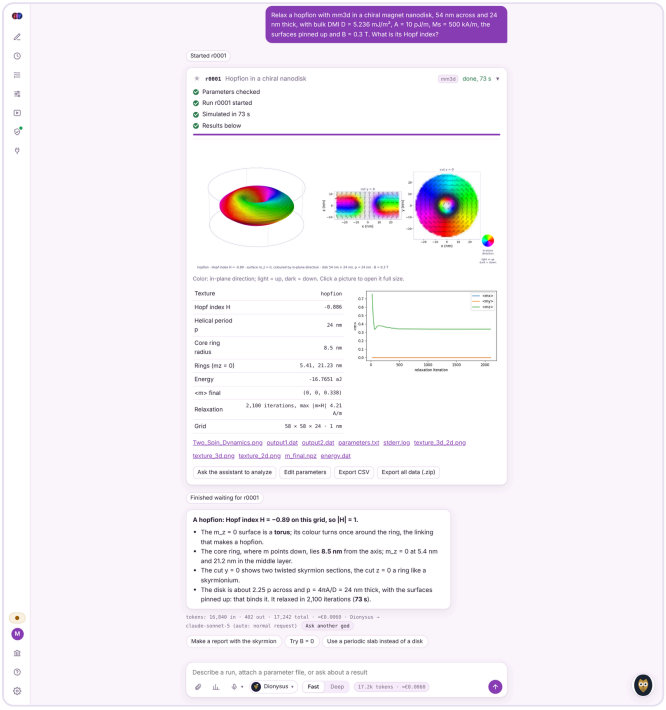
3D micromagnetics: hopfions with mm3d

Choose **mm3d** in the Engine menu, or ask for a hopfion in the chat (it is also a card under *Find a skill*). A hopfion is a 3D soliton: the surface where $m_z = 0$ is a torus, and the in-plane direction turns once around it, so every pair of preimages is linked once; the Hopf index H counts that linking. It is bound in a chiral magnet nanodisk about 2p across and about p thick ($p = 4\pi A/D$) with the surfaces pinned up, at small fields. In a periodic slab it unwinds or grows into helices, and Check warns about it.

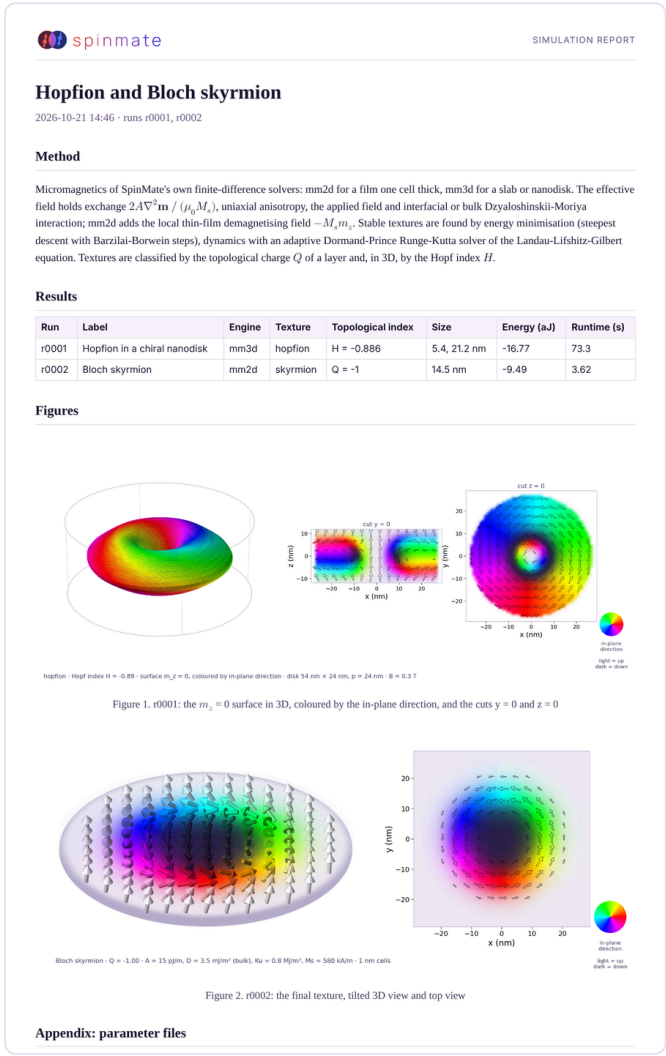
```
--texture=hopfion
--Aex=10e-12
--D=5.236e-3
--Nz=24
--geometry=disk
--disk_diameter=54e-9
--B_ext=0,0,0.3
```



A relaxed hopfion in a 54 nm × 24 nm chiral nanodisk: the $m_z = 0$ torus and the cuts y = 0 and z = 0. H = −0.89 on the grid ($|H| = 1$), core ring at 8.5 nm.



In the chat: a hopfion from a question.

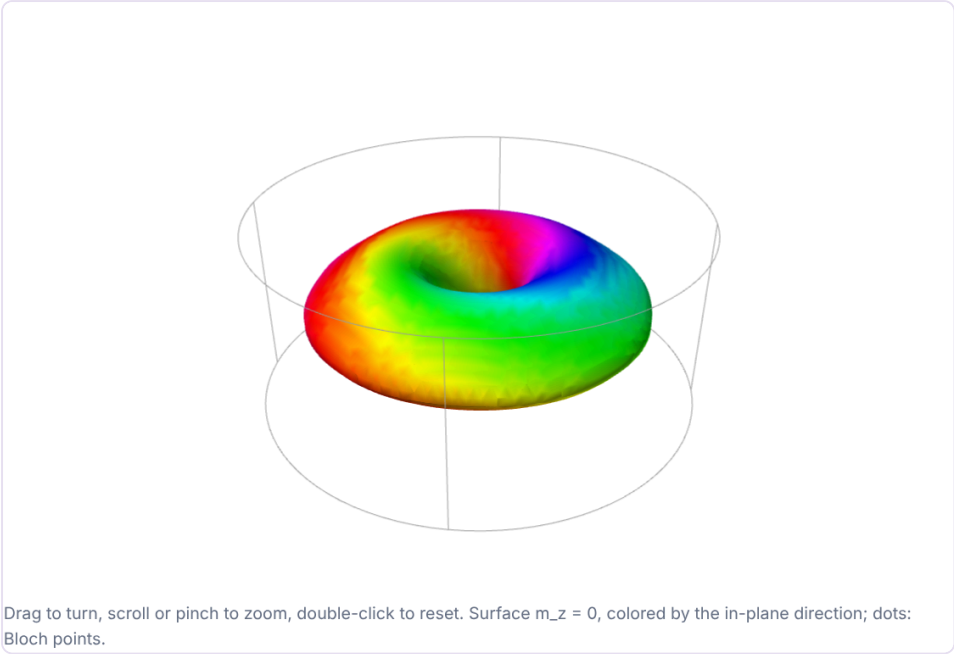


A report of a hopfion and a Bloch skyrmion: the texture pictures are in it.

mm3d check	Expected	Measured
Helix with q = D/2A (bulk DMI)	$-D^2/4A$	within 2 % (exact on the lattice)
One layer against mm2d	same fields	identical

Hopfion through relaxation	$\lvert H\rvert$ stays ≈ 1	$\lvert H\rvert$ = 0.89 (grid)

Interactive 3D view. The run card of an mm3d run opens with the $m_z = 0$ surface drawn in WebGL: drag to turn it, scroll or pinch to zoom, double-click to go back to the first view. The static pictures with the cuts stay below it.

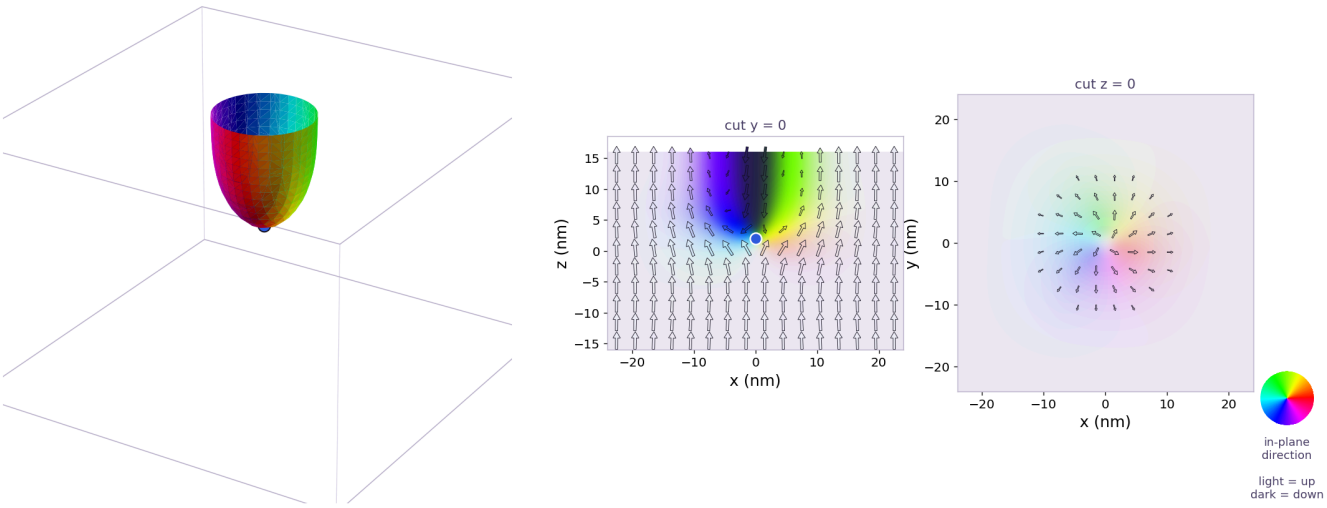


The interactive view of a hopfion on the run card.

Chiral bobbers, torons, H = 2 and Bloch points

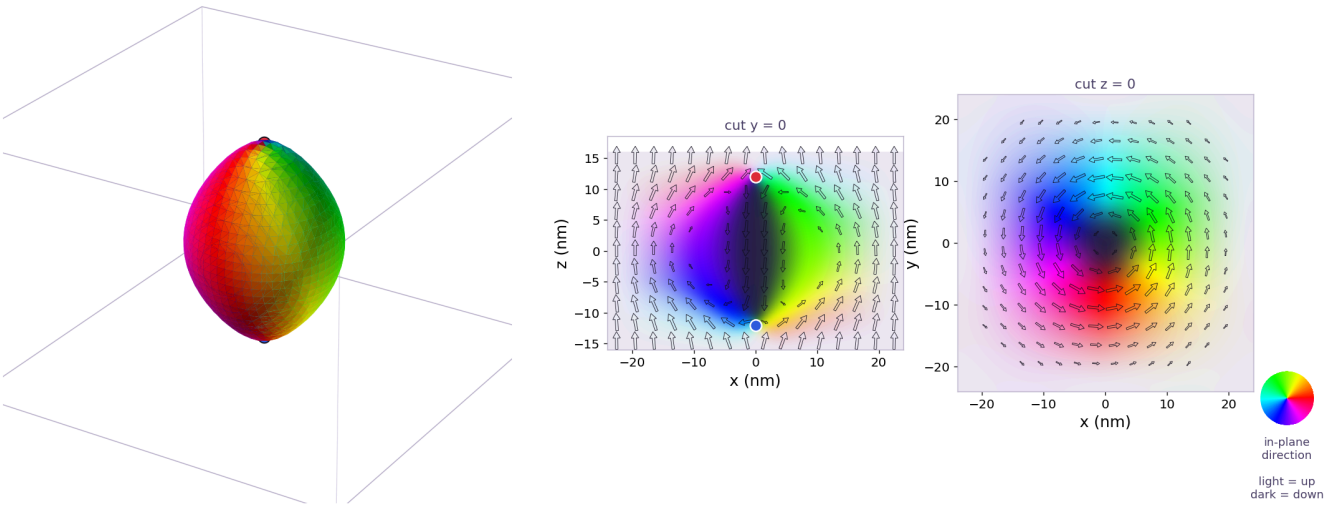
A Bloch point is where m vanishes in the continuum: the solid angle of m over a small closed surface around it is ±4π. mm3d computes that solid angle over every cube of eight cells, so each Bloch point is found with its charge, listed in the summary and drawn as a dot (red +1, blue −1). `texture=chiral_bobber` needs free surfaces and a field between about 0.35 and 0.5 B_D ($B_D = D^2/(A \cdot M_s)$, 5.5 T for the defaults): the tube enters from the top surface and ends about half a period down at a Bloch point. `texture=toron` with pinned surfaces is a tube closed at both ends by Bloch points of opposite charge; it shrinks as the field grows. `texture=hopfion2` stacks two hopfions in a disk about 2p thick, H = 2; a single torus with a double twist unwinds to H = 1 in this model.

```
--texture=chiral_bobber
--geometry=slab
--surfaces=free
--Nx=48
--Ny=48
--Nz=32
--B_ext=0,0,2.2
```



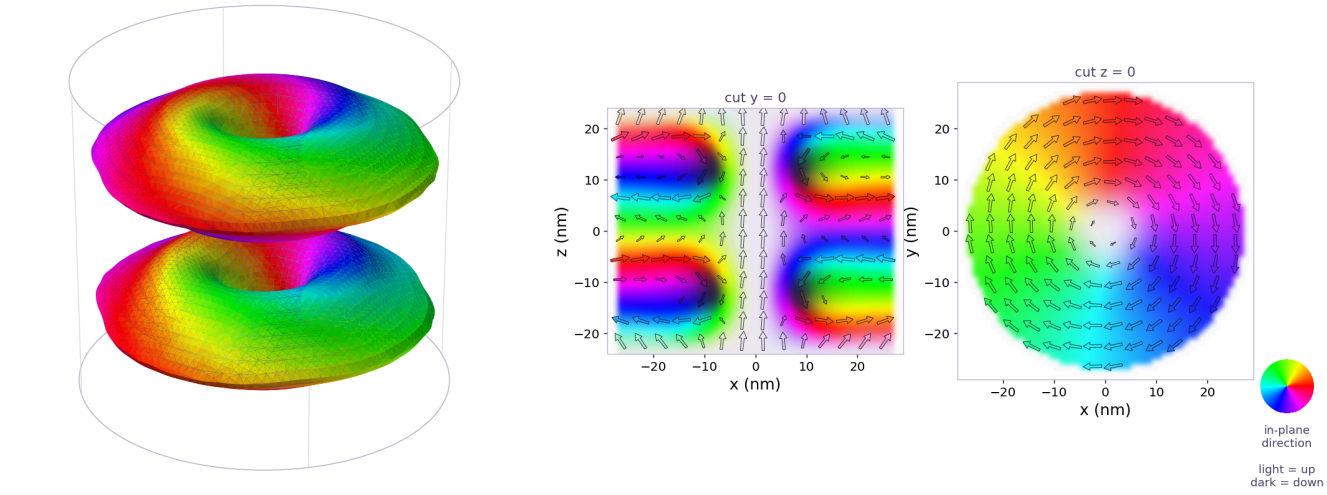
chiral bobber · 1 Bloch point (red +1, blue −1) · surface m_z = 0, coloured by in-plane direction · slab 32 nm thick, p = 24 nm · B = 2.2 T

A chiral bobber at 2.2 T: the tube ends 14 nm below the free top surface at a Bloch point (Q = −1 in the upper layers, 0 below).



toron · 2 Bloch points (red +1, blue −1) · surface m_z = 0, coloured by in-plane direction · slab 32 nm thick, p = 24 nm · B = 1.5 T

A toron between pinned surfaces at 1.5 T: Bloch points +1 and −1 at z = ±12 nm, radius 9 nm in the middle.



hopfion2 · Hopf index H = -1.72 · surface m_z = 0, coloured by in-plane direction · disk 54 nm × 48 nm, p = 24 nm · B = 0.3 T

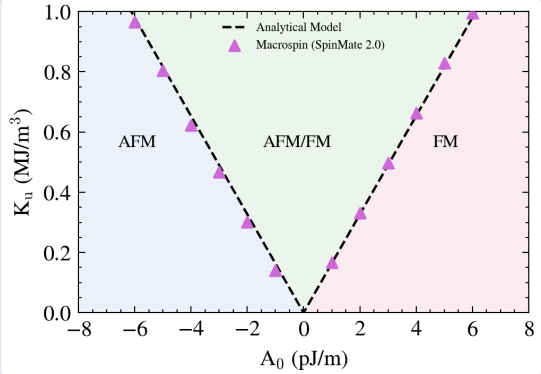
Two hopfions stacked in a 54 nm × 48 nm disk: H = −1.72 on the grid (2 × 0.86).

3D texture check	Expected	Measured

Hedgehog, anti-hedgehog	Bloch point charge ±1	±1, at the centre
Hopfion, skyrmion tube	no Bloch points	none
Chiral bobber	one Bloch point, about p/2 deep	one, 0.5–0.6 p
Toron	Bloch points +1 and −1	+1 above, −1 below
Toron radius, 1 / 1.5 / 2 T	shrinks with field	11.8 / 9.0 / 6.5 nm
hopfion2	H = 2 × H of one hopfion	−1.72 = 2 × −0.86

10 Physics validation

`python -m spinmate_core.validation` runs the real solver on cases with an analytic answer and writes a report (agents read it as `spinmate://validation`). It runs on every pull request that changes the solver or the core. The shield in the left bar is green when every case passes; click it for the report and the phase diagram. Each sublattice carries Ms/2, so $\mu_0 H_E = 4|A0|/(Ms\ l^2)$ and $\mu_0 H_K = 4Ku/Ms$.



AFM/FM phase diagram: the boundary found by bisection with the solver lies on $|Ku| = 2|A0|/l^2$ within 2–6 %.

Case	Expected	Measured
$\|m_1\| = \|m_2\| = 1$	0	2e-16
Same input, same trajectory	0	0
Energy, α = 0, 0.5 ns	conserved	drift 1.6e-6
Kittel (FM mode)	5.62 GHz, 27.99 GHz/T	5.61 GHz, 28.03 GHz/T
AFM resonance	67.3 GHz	67.3 GHz
Time-step convergence	h = 1e-13 ≈ 2e-14 s	0.03 % apart
Gilbert damping, rate α·ω	0.3525 /ns	0.3524 /ns
AFM/FM boundary	$\|Ku\| = 2\|A0\|/l^2$	as expected at ±10 %
Thermal field	isotropic, σ = Temp_Ampl	σ within 3 %

mm2d and mm3d. The same suite checks the micromagnetic engines, and the shield counts all 19 cases:

Case	Expected	Measured
Domain wall, interfacial DMI	$4\sqrt{(A \cdot K_{eff})} - \pi\|D\| = 5.603$ mJ/m², Néel	5.603 mJ/m², Néel
Kittel, perpendicular film	70.78 GHz	70.78 GHz
Spin-wave dispersion f(k)	$\gamma_0/2\pi\ (H_{K,eff} + 2Ak^2/\mu_0 M_s)$	within 2.2 %
STT with β = α (Thiele)	v = u = 50 m/s	49.8 m/s
Tilt at a free edge with DMI	sin θ = Dδ/2A = 0.5889	0.5887
FFT demag: platelet, Néel-wall stray energy	0.8600; 0.0466 mJ/m²	0.8600; 0.0464 mJ/m²

CPU backend vs NumPy	same dm/dt	6e-16 apart
Helix (mm3d)	$-D^2/4A$	within 2 %
Hopfion in a nanodisk (mm3d)	$\|H\| \approx 1$	−0.81 on the grid
Bloch points (mm3d)	±1	+1, −1

Fixed in 2.0. The first run found four issues in `macrospin/`: the static anisotropy was added twice (F1), the Ku(t) amplitude used g_o instead of μ_o (F4), the optional RK4 integrator had wrong weights (F2) and the thermal field was not Gaussian (F3). Runs at T = 0 without Ku(t) give bit-identical trajectories to version 1.0; runs with temperature change.

11 Sweeps, threshold searches and continuing runs

- **Sweep:** “Sweep A0 from −0.25e-12 to +0.25e-12 in 11 steps, 0.5 ns each.” Runs go in parallel; the card shows a table and an overlay.
- **Form:** the chart icon next to the paperclip opens a sweep or threshold search form: parameter in convenient units, range, steps or what to look for, and duration. It writes the request with SI values and warns about unusual numbers.
- **Threshold search:** “Find the smallest SOT current between 50e6 and 150e6 A/cm² that makes the Néel vector rotate.” A few rounds of bracketing sweeps narrow the interval.
- **Continue:** “Continue r0003 for another 0.5 ns.” Time does not restart; *combine* joins the segments into one curve.

12 Comparing runs, figures and reports

- **Compare:** lists the parameters that differ and overlays the curves.
- **Figures:** “Make an IEEE figure of n_z for r0001 and r0003, first 60 ps.” Styles science, IEEE, Nature; PDF, SVG and PNG downloads; *Change…* to adjust.
- **Reports:** SpinMate header, method, results, parameters, figures and appendix, with LaTeX math; print to PDF or download the Markdown version.

13 The Runs panel and questions across runs

The list icon opens every run: search by id, label, #tag or parameter; filter by status and end state; star and tag; select several to compare, plot, report or delete. With the catalog the assistant answers questions across runs in one read-only query:

> *Make me a table of all finished runs: id, A0, initial and final order, first n_z switch in ps.*

14 Settings and conversations

- **Language:** English (US or UK), Greek, Spanish, German or Italian; the buttons, Athena and the answers follow it. It starts from the browser language.
- **API keys:** under each key a check mark and the number of models it can use, or why it does not work. One working key is enough; a key that fails is not used. The × deletes a key from the tab (keys in `.env` are changed there).
- **Theme:** Spin (default), Light or Dark. **Default parameters:** saved per user and used as the base of every run.
- **Cost:** tokens and the estimated € next to the mode buttons; click it for **usage and cost** per day and per provider (Claude, OpenAI, Gemini).

- **Keyboard:** Ctrl+K new conversation, Ctrl+/ the Runs panel, Enter send, Shift+Enter new line, Esc closes a window (Cmd on a Mac).
- **Ask another god:** under an answer, sends the same question to another model level and shows both answers side by side.
- **Conversations:** the clock icon lists your chats; select one, several or all and delete them (runs are kept).
- **Voice:** the microphone records a request; the arrow next to it chooses the microphone and language.

15 Data, privacy and security

- The app listens on this computer only (`BIND_ADDRESS=127.0.0.1`). There is no password; expose it only behind a proxy with real authentication. The MCP endpoint `/mcp` uses the same port and the same rule.
- API keys stay in `.env` (never committed) or in the browser tab. Calls are billed to your own provider account; set `TOKEN_BUDGET` and a monthly limit in the provider console.
- Accounts (name and email) are stored only in `/data/accounts.json` on your machine; nothing is sent to the authors.
- The assistant reads the catalog with read-only SQL and a timeout, and starts runs only through the solver's parser. Over MCP, large requests run only after you agree (`confirmed=true`); in the chat only the **Run it** button approves them.

16 Configuration (`.env`)

Variable	Purpose
`ANTHROPIC_API_KEY` , `OPENAI_API_KEY` , `GEMINI_API_KEY`	Server-side keys (optional; can be pasted in Settings)
`HOST_PORT` , `BIND_ADDRESS`	Port (8421) and who may open the app (127.0.0.1)
`TOKEN_BUDGET`	Token limit per server session
`CONFIRM_MINUTES` , `CONFIRM_RUNS`	When a request waits for **Run it**
`MAX_PARALLEL` , `MAX_SWEEP_RUNS`	Parallel runs and the largest sweep
`DATABASE_URL` , `POSTGRES_PASSWORD`	Run catalog; empty URL switches it off
`EUR_PER_USD` , `PRICES_JSON`	Fixed exchange rate (default: daily ECB) and price overrides
`SMTP_HOST` , `SMTP_PORT` , `SMTP_USER` , `SMTP_PASSWORD` , `SMTP_FROM`	Mail server for the optional welcome email (Gmail: smtp.gmail.com, 587, App Password)
`ADMIN_TOKEN` , `ADMIN_SHEET_URL`	Turns on `/admin` (accounts, welcome emails, outbox, cost of this server) and links the SpinMate users Sheet
`APP_URL`	Base of the links in MCP results (default `http://localhost:8421`)
`MUMAX3_BIN`	Path of the mumax3 binary, for the mumax3 engine
`SSL_CERT_FILE`	Company CA bundle behind a corporate proxy

17 Parameters (most used)

Parameter	Meaning
`--t`	Duration (s)
`--A0`	Inter-sublattice exchange (J/m); negative couples antiparallel
`--m1`, `--m2`	Initial unit vectors
`--SOT_DC_Amp`, `--SOT_pol`	DC spin-orbit current (A/cm²) and its polarisation
`--flagTempVarying`, `--Temp`, `--Fr`	FeRh temperature cycle: on, amplitude (K), frequency (Hz)

A parameter file is plain text, one `--name=value` per line; attach it with the paper clip. Ask "Which parameters can I set?" for the full list with units. Examples are in `examples/`.

```
--t=5e-11
--A0=-0.248e-12
--m1=0.25,0.0,0.968
--m2=0.35,0.0,0.937
```

18 Troubleshooting

Symptom	What to do
"Add an API key"	Paste a key in Settings, or set it in `.env` and run `docker compose up -d --force-recreate`.
No answer, 401 / 429 errors	Check the key and the provider credit; on Auto the next model by score is tried.
New features do not appear	`git pull`, `docker compose up --build -d --force-recreate`, then Ctrl+Shift+R in the browser.
No welcome email	Check `SMTP_*` in `.env`; `/api/config` must show `"email":true`. The guide is always at `/api/guide.pdf`.
Page does not open from another computer	By design; see `BIND_ADDRESS` and chapter 15.
Claude Code does not see SpinMate	Start the Docker app, then `claude mcp list`; the server must show as connected. Restart Claude Code after `claude mcp add`.
"mumax3 not found"	Install mumax3 on a machine with an NVIDIA GPU and set `MUMAX3_BIN`; macrospin runs need nothing extra.
Microphone not listed	Allow the microphone once; open the app on localhost or https.

More in the README on GitHub. Issues and feedback are welcome.